

The schematic diagram illustrates the wiring for the ESP32-C3-WROOM-02 module. Key components and connections include:

- Power Supply:** A +3V3 supply connected to EN (pin 2) via R1 (10k). GND pins are connected to various ground points.
- Resistors:** R1 (10k), R2 (10k), R3 (10k), R4 (100k), R5 (10k), R16 (220R), R31 (10k), and R32 (18k).
- Capacitors:** C1 (1uF), C2 (10uF), and C3 (100nF).
- Switches:** SW1 (RESET), SW2 (BOOT), and SW4 (WIFI_MODE).
- LEDs:** STATUS LED (D3) and CHG_STAT LED.
- Pin Connections:**
 - EN (2) to +3V3 via R1.
 - IO20/RXD (11) and IO21/TXD (12) to CHG_DET (R31) and CHG_STAT (R32).
 - IO2 (R2) pulled up to +3V3.
 - IO8 (R5) pulled up to +3V3.
 - IO9 (R3) pulled up to +3V3.
 - IO4 (R4) pulled up to +3V3.
 - IO5 (R3) pulled up to +3V3.
 - IO10 (SERVO_EN) pulled up to +3V3.
 - IO18 (USB_DP_MCU) and IO19 (USB_DM_MCU) connected to USB.

Power Flags:

Flag	Pin	Value
PWR_FLAG	IO2	R2
PWR_FLAG	IO8	R5
PWR_FLAG	IO9	R3
PWR_FLAG	IO4	R4
PWR_FLAG	IO5	R3

ESP32-C3 notes (MCU box):

- Strapping: IO2 (R2), IO8 (R5), IO9 (R3) pulled up 10k => SPI boot.
- BOOT (SW2) low at reset => USB download mode.
- IO8 drives the STATUS LED active-low (LED off = strap high).
- IO10 = SERVO_EN; R4 100k keeps the servo rail off at reset/sleep.
- EN: R1 10k + C1 1uF RC delay (Espressif HW design guide).
- USB: IO18/IO19 native USB Serial/JTAG - no UART bridge.

SW4 (05103011MA7QP1, common = pin 2):

- POS1 MODE_A (IO4) low = OFF
- POS2 both high = SPARSE
- POS3 MODE_B (IO5) low = CONTINUOUS.

Internal pull-ups only. IO4/IO5 are deep-sleep wake-capable (GPIO0-5). VBAT_SENSE IO3 / VBUS_SENSE IO1 = ADC1.

Diagram illustrating the connections for the servo motor:

- GND connected to PWR_FLAG
- VBUS connected to PWR_FLAG
- VSYS connected to PWR_FLAG
- VBAT connected to PWR_FLAG
- SERVO_PWR connected to PWR_FLAG

ESP32-C3 notes (MCU box):
 Strapping: IO2 (R2), IO8 (R5), IO9 (R3) pulled up 10k → SPI boot.
 BOOT (SW2) low at reset → USB download mode.
 IO8 drives the STATUS LED active-low (LED off = strap high).
 IO10 = SERVO_EN; R4 100k keeps the servo rail off at reset/sleep.
 EN; R1 10k + C1 1uF RC delay (Expressif HW design guide).
 USB: IO18/IO19 native USB Serial/JTAG – no UART bridge.

SW4 (05103011MA7QP1, common = pin 2):
 POS1 MODE_A (I04) low = OFF | POS2 both high = SPARSE | POS3 MODE_B (I05) low = CONTINUOUS.
 Internal pull-ups only. I04/I05 are deep-sleep wake-capable (GPIO0-5). VBAT_SENSE I03 / VBUS_SENSE I01 = ADC1.

USB-C Input + USB data + ESD + VBUS sense

The diagram illustrates a circuit for a USB-C connector (J1) that supports VBUS sensing, data communication, and ESD protection. The connector pins are connected as follows:

- VBUS:** Connected to pin A4. The VBUS line passes through a 47.5k resistor (R12) and is sensed at VBUS_SENSE.
- Data Lines (D+, D-):** Connected to pins A6, B6 (D+) and A7, B7 (D-). These lines pass through 5.1k resistors (R10, R11) to GND. They are also connected to the USB-D- and USB-D+ pins of the ESD diode (U6).
- ESD Protection:** The ESD diode (U6, USBLC6-2SC64.7uF) is connected to GND and the VBUS line. It also provides protection for the data lines.
- MCU Interface:** The USB_DM_MCU and USB_DP_MCU pins of the MCU are connected to the USB-D- and USB-D+ pins of the ESD diode, respectively, through 22R resistors (R25, R24).
- Shielding:** The shield is connected to the SH pin and the A1 pin is connected to GND.

LiPo Charger (MCP73831) + power switch

Power switch: SW3 (0.1 A) only drives Q5's gate.
 POS1 (1-2): gate = GND -> Q5 on
 POS2 (2-3): gate = source -> Q5 off
 R26 10M holds Q5 off in the break-before-make gap.

Power switch: SW3 (0.1 A) only drives Q5's gate.
 POS1 (1-2): gate = GND -> Q5 on
 POS2 (2-3): gate = source -> Q5 off
 R26 10M holds Q5 off in the break-before-make gap.

Servo Output

The diagram illustrates the wiring for a servo motor output. The SERVO_PWM signal is connected to the SERVO_SIG pin (pin 1) of the servo connector (J3) through a 100R resistor (R17). The SERVO_PWR signal is connected to the SERVO_PWR pin (pin 2) of the connector. A 100uF capacitor (C13) is connected between the SERVO_PWR line and ground (GND). The servo motor is labeled SERVO.

Air Sensor (BME688)

Servo rail load switch (v8: no boost, the servo runs from VSYS = USB via D1 ~4.5 V or the battery 3.0–4.2 V).
Q6 connects SERVO_PWR to VSYS only while SERVO_EN (IO10) is high.
Q7 pulls the gate to ~VSYS–2.5 V via R28; R4 keeps it off at reset/sleep.
C22 (gate–drain) slews SERVO_PWR at ~0.5 V/ms → ~60 mA inrush into C13 + C19 (122 uF).
Firmware: SERVO_EN high ≥ 50 ms before a move; skip moves below VBAT ~3.5 V.

Servo rail load switch (v8: no boost, the servo runs from VSYS = USB via D1 -4.5 V or the battery 3.0-4.2 V). Q6 connects SERVO_PWR to VSYS only while SERVO_EN (I010) is high. Q7 pulls the gate to -VSYS-2.5 V via R28; R4 keeps it off at reset/sleep. C22 (gate-drain) slews SERVO_PWR at -0.5 V/ms -> -60 mA inrush into C13 + C19 (122 uF). Firmware: SERVO_EN high >= 50 ms before a move: skip moves below VBAT-3.5 V.

The schematic diagram illustrates the internal wiring of the STEMMA QT port (v7). It shows the connection of the USBLC6-25C6 module (U8) to the I2C bus. Key components include:

- Q8 (DMG2305UX):** A MOSFET used for switching the I2C bus from IO2 (low) when the device is in reset or sleep.
- Q9/Q10 (BSS138):** NPN transistors used to isolate the bus when the device is off.
- R29/R30 (10k):** Pull-up resistors for the SCL and SDA lines.
- C23/C24 (100nF):** Decoupling capacitors for the I2C bus.
- U8 (USBLC6-25C6):** The USB-to-I2C bridge module.
- J5 (STEMMA QT port):** The 4-pin connector for the STEMMA QT module.

The diagram shows the connection of the STEMMA QT port (J5) to the USBLC6-25C6 module (U8) and the I2C bus lines (SCL, SDA). The USBLC6-25C6 module is connected to the I2C bus via its SDA and SCL pins. The STEMMA QT port (J5) is connected to the USBLC6-25C6 module via its 4 pins (1 GND, 2 3V3, 3 SDA, 4 SCL).

STEMMA QT port (v7): J5 1 GND, 2 3V3, 3 SDA, 4 SCL.
Q8 switches it from I02 (low = on; R2 keeps it off at reset/sleep).
Q9/Q10 isolate the bus when off, R29/R30 port pull-ups, U8 ESD.
BME688 now at 0x76 (SDO = GND) - 0x77 free for add-ons.
R31/R32: change STAT (5 V) → I020 CHG_DET. Low = charging (with VBUS), high = done.

ESP32-C3 + BME688 + SG92R (v7)

Sheet: /
File: air-quality-pcb-v9.kicad_sch

Title: Air Quality Checker

Size: A3	Date:
KiCad E.D.A. 10.0.6	

Rev: I
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